

Anirban Ray

Final-year PhD Student in Computer Science

Computational Image Restoration | Generative AI for Microscopy

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Profile

Final-year Computer Science PhD student developing generative AI and deep learning methods for computational microscopy. Research focus: inverse problems, image restoration, dehazing, super-resolution, denoising, uncertainty estimation, and posterior sampling. Prior industry experience as an AI researcher at Hitachi R&D in Tokyo.

Education

Doctorate (Dr. rer. nat.) in Computer Science **2022–2026 (expected)**

Technische Universität Dresden & Human Technopole

Dresden, Germany / Milan, Italy

- Advisor: Dr. Florian Jug; Thesis Advisory Committee: Prof. Dr. Ivo F. Sbalzarini and Dr. Virginie Uhlmann.
- Thesis: *Computational Image Restoration with Conditional Generative Models for Microscopy*.
- Selected outputs: ResMatching (IEEE ISBI 2026 oral), HazeMatching (CVPR 2026 Findings), and RESOLFT deep-learning imaging (Nature Communications, 2026).

Master of Engineering in Computer Science

2016–2018

Nagoya Institute of Technology

Nagoya, Japan

- Advisors: Prof. Jun Sato and Prof. Fumihiko Sakaue.
- Thesis: *Modeling the Feature Evolution in CNNs using LSTM*; focus on computer vision and deep learning.

Research Student

Oct. 2015–Mar. 2016

Nagoya Institute of Technology

Nagoya, Japan

Bachelor of Technology in Computer Science and Engineering

2011–2015

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Chennai, India

Research Experience

Time-lapse RESOLFT nanoscopy empowered by deep learning

2024–2025

Human Technopole; collaboration with the Testa Lab

Milan, Italy

- Applied deep learning restoration to low-SNR, sub-sampled RESOLFT nanoscopy, enabling $5\times$ longer imaging with $10\times$ lower light dose or $4\times$ faster imaging while preserving approximately 60 nm resolution.
- Output: Nature Communications, 2026 (in Press); [Research Square preprint](#).

ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching

2025

Human Technopole

Milan, Italy

- Developed guided conditional flow matching for microscopy super-resolution, unifying denoising, uncertainty estimation, and posterior sampling.
- IEEE ISBI 2026 oral presentation; [GitHub](#), [arXiv](#), [project page](#).

HazeMatching: Conditional Flow Matching for Microscopy Dehazing

2023–2025

Human Technopole

Milan, Italy

- Created a conditional flow-matching framework for microscopy dehazing, widefield-to-confocal restoration, and Expansion Microscopy super-resolution.
- CVPR 2026 Findings; CV4Science Workshop poster; [GitHub](#), [arXiv](#).

Deep Learning for Microscopy Image Analysis **2018–2021**
Hitachi Ltd., Center for Technology Innovation *Tokyo, Japan*

- Built deep learning systems for biomedical microscopy and SEM image analysis, including segmentation, counting, classification, quantification, CISS publications, and granted patents.

Modeling the Feature Evolution in CNNs using LSTM **2016–2018**
Nagoya Institute of Technology *Nagoya, Japan*

- Modeled feature evolution across CNN layers using LSTM networks for image classification.

Professional Experience

Associate Researcher **Apr. 2018–Jan. 2022**
Hitachi Ltd., R&D Group, Center for Technology Innovation *Tokyo, Japan*

- Delivered deep learning and computer vision systems for industrial R&D and biomedical microscopy image analysis.

Engineering Intern, AR Applications **Sept. 2016**
Sun Corporation *Konan, Japan*

Engineering Intern, Cloud Services **Aug. 2014–Jan. 2015**
Machine Pulse (Mahindra Tego) *Mumbai, India*

Publications and Preprints

- [1] **Anirban Ray et al.**. [ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching](#). *IEEE ISBI 2026*. Accepted; oral presentation.
- [2] **Anirban Ray et al.**. [HazeMatching: Conditional Flow Matching for Microscopy Dehazing](#). *CVPR 2026 (Findings)*. Accepted; CV4Science Workshop poster.
- [3] **Minet, Ray et al.**. [Time-lapse RESOLFT nanoscopy empowered by deep learning](#). *Nature Communications 2026 (in Press)*.
- [4] **Kakishita, Ray et al.**. [Quantitative Analysis System for Bacterial Cells in SEM Image using Deep Learning](#). *CISS 2021*. pp. 1–6.
- [5] **Kakishita, Ray et al.**. [Deep Learning Based Bacteria Classification from SEM Images Using a Combination of Membrane and Internal Features](#). *CISS 2022*. pp. 13–18.

Patents

- Inventor on 3 granted Hitachi High-Tech image-processing patents: [US12327363B2](#), [US12211213B2](#), and [EP3961562B1](#).

Presentations and Posters

- Invited talk:** Learning to Recover Biological Structure under Uncertainty in Light Microscopy. AlQuraishi Laboratory, Department of Systems Biology, Columbia University, New York City, USA, Oct. 2026.
- Invited talk:** Scientific Inverse Problems in the Age of Generative AI. AI & Advanced Computing Institute, Schmidt Sciences, New York City, USA, Jun. 2026.
- Talk and poster:** [ResMatching](#). IEEE ISBI 2026, London, UK, Apr. 2026.
- Poster:** [HazeMatching](#). IEEE/CVF CVPR 2026 and CV4Science Workshop, Denver, CO, USA, Jun. 2026.
- Invited talk:** Introduction to Diffusion Model for Microscopy. EMBO-DL4MIA, Milan, Italy, May 2024.

Teaching and Mentoring

Teaching Assistant, Deep Learning for Microscopy Image Analysis (DL4MIA) **2022, 2023, 2024**
Human Technopole *Milan, Italy*

- Mentored participants, supported coding sessions and projects, and delivered an invited lecture in the 2024 edition.

Teaching Assistant, Deep Learning at MBL (DL@MBL)

2023, 2024

Marine Biological Laboratory

Woods Hole, USA

- Led hands-on support for deep learning in biological imaging; co-mentored project work linked to NeurIPS 2024 and RESOLFT microscopy outputs.

Supervision of summer intern

2026

Human Technopole

Milan, Italy

- Supervised one undergraduate intern during my PhD for two months (summer internship project), providing research guidance, technical mentoring, and feedback on project direction.

Professional Service

- Technical Program Committee Reviewer, *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2027
- Reviewer, [AI4Life Open Call](#), 2024.

Awards, Honors, and Selected Training

- **Selected Participant, Optical Microscopy & Imaging in the Biomedical Sciences (OMIBS)**, Marine Biological Laboratory, USA, 2023; team won 3rd place in microscopy imaging.
- **Selected Participant, International Computer Vision Summer School (ICVSS)**, Italy, 2017; selected among 150 global students.
- **Aichi Monozukuri Scholarship**, Aichi Prefectural Government, Japan, 2015; full funding for master's studies, selected among top 10 students in Asia.
- **Selected Participant, Undergraduate Summer School on Computer Science**, Indian Institute of Science, Bengaluru, India, 2013.

Projects and Software

- **ResMatching**: Open-source implementation, data, models, project page, and quickstart notebook for guided conditional flow matching in microscopy super-resolution.
- **HazeMatching**: Open-source implementation, data, models, and quickstart notebook for conditional flow matching in microscopy dehazing and Expansion Microscopy super-resolution.

Skills

Research

Computational microscopy; image restoration; inverse problems; generative AI; conditional flow matching; diffusion models; denoising; dehazing; super-resolution; uncertainty estimation; posterior sampling; segmentation; computer vision.

Programming Languages

Python; PyTorch; OpenCV; NumPy; Bash; Linux; Git; SLURM; L^AT_EX.
Bengali (native); Hindi (fluent); English (fluent); Japanese (conversational); Tamil (elementary); Italian (elementary).